



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES



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COURSE: CSA0915 -JAVA PROGRAMMING

SLOT: B

SCENARIO BASED QUESTIONS

1. A teacher has stored the marks of 10 students in an array. She wants to analyze the performance of the class by finding the highest mark, calculating the average mark, and counting how many students scored above the average.

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JAVA PROGRAMMING

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```
1 class StudentMarksAnalysis
2 {
3     public static void main(String[] args)
4     {
5
6         int[] marks = {45, 67, 89, 32, 76, 50, 91, 84, 60, 73};
7
8         int highest = marks[0];
9         int sum = 0;
10
11         for (int i = 0; i < marks.length; i++)
12         {
13             sum += marks[i];
14
15             if (marks[i] > highest)
16             {
17                 highest = marks[i];
18             }
19         }
20
21         double average = (double) sum / marks.length;
22
23         int count = 0;
24         for (int i = 0; i < marks.length; i++)
25         {
26             if (marks[i] > average)
27             {
28                 count++;
29             }
30         }
31         System.out.println("Highest = " + highest);
32         System.out.printf("Average = %.1f\n", average);
33         System.out.println("Students Above Average = " + count);
34     }
35 }
```

10, 20, 30, 40, 50, 60, 70, 80, 90, 100

Output

Highest = 91
Average = 66.7
Students Above Average = 6

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

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- JAVA PROGRAMMING
- Your Code
- ```
1 import java.util.*;
2
3 class Attendance {
4 public static void main(String[] args) {
5 Scanner sc = new Scanner(System.in);
6 String attendance = sc.nextLine();
7 int present = 0;
8 int absent = 0;
9
10 for (int i = 0; i < attendance.length(); i++) {
11 char ch = attendance.charAt(i);
12
13 if (ch == 'P') {
14 present++;
15 } else if (ch == 'A') {
16 absent++;
17 }
18 }
19
20 int percentage = (present * 100) / attendance.length();
21
22 System.out.println("Present = " + present);
23 System.out.println("Absent = " + absent);
24 System.out.println("Attendance = " + percentage + "%");
25
26 if (percentage < 75) {
27 System.out.println("Not Eligible");
28 } else {
29 System.out.println("Eligible");
30 }
31 }
32 }
33 }
```
- Input
- PPPPPPPPPP
- Output
- ```
Present = 10
Absent = 0
Attendance = 100%
Eligible
```
- Visible Test Cases
- Test Case 1
-
- Test Case 2
-
- Test Case 3
-

Your Code

```
1 class StudentGrades
2 {
3     public static void main(String[] args) {
4
5         int[] marks = {45, 67, 89, 32, 76, 50};
6
7         int pass = 0;
8         int fail = 0;
9
10        for (int i = 0; i < marks.length; i++) {
11
12            if (marks[i] >= 50) {
13                pass++;
14            } else {
15                fail++;
16            }
17
18            if (marks[i] >= 80) {
19                System.out.println(marks[i] + " - Grade A");
20            } else if (marks[i] >= 60) {
21                System.out.println(marks[i] + " - Grade B");
22            } else if (marks[i] >= 50) {
23                System.out.println(marks[i] + " - Grade C");
24            } else {
25                System.out.println(marks[i] + " - Fail");
26            }
27        }
28
29        System.out.println("Pass = " + pass);
30        System.out.println("Fail = " + fail);
31    }
32 }
```

Input

stdin input

Output

45 - Fail
67 - Grade B
89 - Grade A
32 - Fail
76 - Grade B
50 - Grade C

Visible Test Cases

Test Case 1
Test Case 2
Test Case 3

- An organization maintains employee names in an array. Identify duplicate names and count occurrences.

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JAVA PROGRAMMING

```
1 import java.util.*;
2 class DuplicateNames {
3     public static void main(String[] args) {
4
5         String[] names = {"Ram", "Sita", "Ram", "John", "Sita"};
6
7         for (int i = 0; i < names.length; i++) {
8             int count = 1;
9             boolean duplicate = false;
10
11             for (int j = 0; j < i; j++) {
12                 if (names[i].equals(names[j])) {
13                     duplicate = true;
14                     break;
15                 }
16             }
17
18             if (!duplicate) {
19                 for (int j = i + 1; j < names.length; j++) {
20                     if (names[i].equals(names[j])) {
21                         count++;
22                     }
23                 }
24
25                 if (count > 1) {
26                     System.out.println(names[i] + " = " + count);
27                 }
28             }
29         }
30     }
31 }
```

stdin input

Output

Ram = 2
Sita = 2

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Run Save

5. A bank stores transactions where positive values are deposits and negative values are withdrawals. Calculate final balance and count deposits and withdrawals.

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JAVA PROGRAMMING

```
1 import java.util.*;
2 class BankTransactions {
3     public static void main(String[] args) {
4
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter number of transactions: ");
8         int n = sc.nextInt();
9
10        int[] transactions = new int[n];
11
12        System.out.println("Enter the transactions:");
13
14        for (int i = 0; i < n; i++) {
15            transactions[i] = sc.nextInt();
16        }
17
18        int balance = 0;
19        int deposits = 0;
20        int withdrawals = 0;
21
22        for (int i = 0; i < n; i++) {
23
24            balance = balance + transactions[i];
25
26            if (transactions[i] > 0) {
27                deposits++;
28            } else if (transactions[i] < 0) {
29                withdrawals++;
30            }
31        }
32
33        System.out.println("Balance = " + balance);
34        System.out.println("Deposits = " + deposits);
35        System.out.println("Withdrawals = " + withdrawals);
36    }
37 }
```